Page 10 Practical Report

Task Tracker Status Filter

Date: 21 October 2026

Objective

Implement a focused status filter for the Task Tracker and verify that it returns the correct tasks for each selection. The implementation also addresses a case-sensitivity bug, protects existing “All” behavior, and includes regression checks for expected and edge-case inputs.

Implementation Details

Area	Work completed
All status	Selecting All returns the complete task list and preserves the original task objects, so existing behavior is not changed.
Open and Done filters	Selecting Open returns only Open tasks, while selecting Done returns only Done tasks. The negative test confirms that Open tasks do not appear in Done results.
Case-insensitive matching	The bug diagnosis identified that a case-sensitive comparison can miss a task when its stored status is written as done but the selected filter is Done. The fix normalizes status values before comparing them.
Edge-case handling	Additional tests check unknown status values, an empty task list, and malformed task status data. Invalid status data is safely ignored rather than causing the filter to crash.
Review and handoff	The project includes a bug-diagnosis note describing the symptom, root cause, focused fix, and scope; a diff of the relevant implementation and tests; the exact Git revision; and a handoff document summarizing verification and limitations.

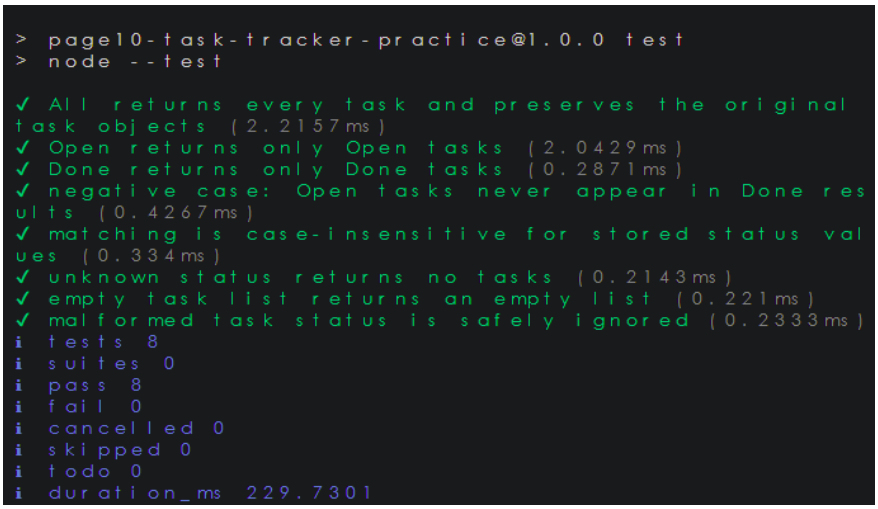
Verification Results

The automated test run completed successfully. All eight checks passed, with no failures, cancellations, or skipped tests.

Total Tests	Passed	Failed	Result
8	8	0	**PASS**

Test Evidence

The terminal screenshot below records the local test command and its results: 8 tests passed and 0 failed.



Conclusion

The Task Tracker status filter has been implemented with the All, Open, and Done options. The completed test run verifies correct filtering, case-insensitive status matching, preservation of existing behavior, and safe handling of invalid or empty inputs. The supporting diagnosis and handoff notes document the change and its verification.